

TLB Impact on Memory Access Time

1. Introduction

Modern computer systems use virtual memory to efficiently manage physical memory. The Translation Lookaside Buffer (TLB) is a cache that stores recent address translations to speed up memory access.

2. Background

When a CPU accesses memory, it checks the TLB first. If the page number is found (TLB Hit), access is fast. If not found (TLB Miss), the system must access the page table in main memory, increasing delay.

3. Problem Statement

To analyze how TLB hit ratio and miss penalty affect Effective Memory Access Time (EMAT).

4. Objectives

- Understand TLB working mechanism.
- Derive and explain EMAT formula.
- Compare performance with and without TLB.
- Perform numerical and graphical analysis.

5. Methodology

- Literature study on virtual memory and TLB.
- Mathematical derivation of EMAT.
- Simulation using programming language.
- Graphical analysis of hit ratio vs EMAT.

6. EMAT Formula

$$EMAT = h(t + m) + (1 - h)(t + 2m)$$

Where h = hit ratio, t = TLB access time, m = memory access time.

7. Expected Outcome

Higher TLB hit ratio significantly reduces memory access time, improving overall system performance.

8. Conclusion

The TLB plays a crucial role in improving system speed by reducing address translation time. Efficient TLB design leads to better computer performance.